

# Paper IV - Recent Advances (MCh Urology, Summer 2015)

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## IV-1. Role of robotics in urology

**Type:** LAQ (25 marks) | ~1500 words

### Introduction / Definition

- **Robot:** a reprogrammable, multifunctional manipulator. Surgical systems in use today are **master-slave telemanipulators**. The surgeon controls every movement, so the system is not autonomous.
- Urology took up robotics earlier and more widely than any other specialty, largely because of **robot-assisted radical prostatectomy (RARP)**. In many high-income countries RARP is now the commonest approach to radical prostatectomy.
- Campbell: Robotic and laparoscopic surgery chapters (Basics of laparoscopic and robotic surgery).

### History (milestones)

| Year        | Milestone                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1985        | <b>PUMA 560:</b> CT-guided stereotactic brain biopsy (neurosurgery)                                                                        |
| Late 1980s  | <b>PROBOT</b> (Imperial College, London): first urological robot, for automated TURP-like prostate resection                               |
| 1990s       | <b>AESOP</b> (Computer Motion): voice-controlled laparoscope holder. <b>ZEUS:</b> master-slave system                                      |
| 2000        | <b>da Vinci</b> (Intuitive Surgical) FDA-cleared for general laparoscopic surgery                                                          |
| 2000        | <b>Binder &amp; Kramer</b> (Frankfurt): first robotic radical prostatectomy                                                                |
| 2001        | <b>"Lindbergh operation"</b> (Marescaux): transatlantic telesurgical cholecystectomy, New York to Strasbourg, using ZEUS                   |
| Early 2000s | <b>Menon</b> (Vattikuti Urology Institute, Detroit): standardised <b>VIP (Vattikuti Institute Prostatectomy)</b> , which made RARP popular |

### da Vinci system: components

1. **Surgeon console:** 3D high-definition stereoscopic viewer, master controllers, foot pedals (camera, clutch, energy), motion scaling and tremor filtration.
2. **Patient-side cart:** 3-4 robotic arms carrying the camera and **EndoWrist** instruments with 7 degrees of freedom.

3. **Vision cart:** image processing, light source, energy generator, assistant monitor.

**Generations:** standard (2000), **S** (2006), **Si** (2009; dual console for training, Firefly fluorescence), **Xi** (2014; boom-mounted rotating arms, thinner arms, laser targeting, multiquadrant access), **SP** (single port, single 2.5 cm cannula; FDA approval for urology 2018), **da Vinci 5** (FDA 2024; force-feedback instruments, greater computing power).

#### Other platforms

- **Hugo RAS** (Medtronic): modular, independent arm carts, open console.
- **Versius** (CMR Surgical, UK): small modular arms, open console.
- **Senhance** (Asensus): haptic feedback, eye-tracking camera control, reusable instruments.
- **SSI Mantra** (SS Innovations, India): Indian low-cost modular system, also used for telesurgery.
- Cheaper competitors are expected to bring costs down, which matters especially for India.

#### Advantages and disadvantages

| Advantages                                                | Disadvantages                                                         |
|-----------------------------------------------------------|-----------------------------------------------------------------------|
| 3D magnified vision                                       | High capital (~US\$1.5–2.5 million) and disposable costs              |
| 7 DOF wristed instruments                                 | Limited haptic feedback (partly addressed in da Vinci 5 and Senhance) |
| Tremor filtration, motion scaling                         | Docking time; large footprint                                         |
| Ergonomic for surgeon                                     | Surgeon remote from patient (emergency undocking)                     |
| Shorter learning curve than pure laparoscopy for suturing | Learning curve still exists; credentialing issues                     |
| Less blood loss, shorter stay than open                   | Port-site hernia; specific positioning complications                  |
| Ideal for reconstruction (intracorporeal suturing)        | Limited availability in public sector                                 |

#### Applications in urology (with evidence)

##### 1. Radical prostatectomy (RARP): the index procedure

- **Yaxley et al., Lancet 2016:** single-surgeon RCT of open retropubic vs robotic prostatectomy. Urinary and sexual function were **similar at 12 weeks**. The robotic arm had less blood loss and a shorter hospital stay.
- **Coughlin et al., Lancet Oncol 2018** (same trial): functional outcomes were still **similar at 24 months**.
- Systematic reviews show lower transfusion rates and shorter stay. Oncological outcomes are comparable to open surgery.
- Technical refinements: Retzius-sparing (Bocciardi) for earlier continence, NeuroSAFE, extended PLND, and single-port/transvesical SP-RARP.

##### 2. Robotic partial nephrectomy (RAPN)

- Compared with laparoscopic partial nephrectomy: **shorter warm ischaemia time, lower conversion to radical nephrectomy**, a shorter learning curve, and the ability to treat complex/hilar tumours.
- Adjuncts include **Firefly (ICG)** for selective clamping, zero ischaemia techniques and 3D virtual models.
- In many centres it has become the preferred minimally invasive approach.

##### 3. Robotic radical cystectomy (RARC)

- **RAZOR trial (Parekh et al., Lancet 2018):** RARC was **non-inferior** to open cystectomy for **2-year progression-free survival**, with less blood loss, fewer transfusions and a slightly shorter stay.

- **iROC trial (Catto et al., JAMA 2022):** RARC with **intracorporeal** diversion increased **days alive and out of hospital at 90 days**, and reduced thromboembolism and wound complications, compared with open surgery.
  - Intracorporeal neobladder: Karolinska (Wiklund) technique.
- 4. Pyeloplasty:** success above 90–95%. Robotic surgery is widely used for adult and paediatric PUJ obstruction and redo pyeloplasty.
- 5. Ureteric reimplantation**
- Adults: Boari flap and psoas hitch for distal strictures.
  - Children: **RALUR** (robot-assisted laparoscopic extravesical ureteral reimplantation, Lich-Gregoir) for VUR, with success rates approaching open surgery in experienced hands.
- 6. Ureteric reconstruction: robotic buccal mucosa graft (BMG) ureteroplasty** for long proximal/mid-ureteric strictures, ureterocalycostomy, ileal ureter, and ureteroureterostomy for retrocaval ureter.
- 7. RPLND (R-RPLND):** primary RPLND for low-volume stage II seminoma or NSGCT, and selected post-chemotherapy cases. Template dissection can preserve ejaculation.
- 8. Adrenalectomy:** equivalent to laparoscopy and helpful in obese patients, partial adrenalectomy and large tumours.
- 9. Robotic simple prostatectomy:** for large glands (>80–100 mL). Less bleeding and shorter stay than open surgery.
- 10. Sacrocolpopexy:** robotic approach for apical prolapse. Also used for VVF/vesicouterine fistula repair, Mitrofanoff and augmentation cystoplasty.
- 11. Robotic kidney transplantation (RAKT):** developed by Menon, Vattikuti and the ERUS-RAKT group (Breda). Lower wound complications and lymphocele rates, useful in obese recipients, with regional hypothermia.
- 12. Other uses:** robotic radical nephrectomy with IVC thrombectomy, nephroureterectomy, ilioinguinal lymph node dissection (VEIL) and diverticulectomy.

## Training and credentialing

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- Structured pathway: didactics, then **virtual-reality simulators** (dV-Skills Simulator, Mimic dV-Trainer, RobotiX), then dry and wet lab, then modular supervised console time.
- **Fundamentals of Robotic Surgery (FRS):** a validated, multispecialty curriculum similar to FLS.
- **ERUS curriculum** (EAU Robotic Urology Section): modular training with certification.
- **Dual console** (Si and Xi) for mentoring. Credentialing is decided by each institution (minimum case numbers, proctoring).
- **Learning curve for RARP:** commonly quoted as many tens of cases for operative proficiency and more than 150–250 for optimal margins and continence.

## Cost

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- Costs come from the capital purchase, annual maintenance and limited-use instruments.
- Robotic surgery is cost-effective only at high volume.
- Cheaper alternatives include new platforms, reusable instruments, multispecialty use and indigenous Indian systems (SSI Mantra).

## Complications specific to robotics

| Problem                             | Details                                                                                                                                                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Steep Trendelenburg</b> (25–45°) | Raised <b>intraocular pressure</b> ; <b>ischaemic optic neuropathy</b> (rare, blindness); corneal abrasion; raised <b>ICP</b> ; facial/laryngeal oedema; reduced FRC and ventilatory difficulty |
| Positioning                         | Brachial plexus injury, <b>lower-limb compartment syndrome / rhabdomyolysis</b> in prolonged lithotomy, peroneal nerve palsy                                                                    |
| Instrument-related                  | Insulation failure causing stray energy and bowel burns; instrument malfunction                                                                                                                 |
| Lack of haptics                     | Tissue tearing, suture breakage                                                                                                                                                                 |
| Others                              | Port-site hernia; CO <sub>2</sub> embolism; conversion; system failure (rare; most cases are completed or converted)                                                                            |

**Prevention:** limit Trendelenburg angle and time, padding, shoulder supports avoided or well padded, and lithotomy kept under about 2–3 h where possible.

## Recent advances

- **Single-port (SP) surgery:** extraperitoneal and transvesical RARP, with the possibility of same-day discharge.
- **Telesurgery:** since 2023–25, long-distance telesurgery (including RARP and nephrectomy) has been reported from China and India (SSI Mantra) over 5G and fibre networks. Latency remains the main limitation.
- **Haptic / force feedback:** da Vinci 5.
- **Image guidance and augmented reality:** 3D-printed and virtual models overlaid onto the console view; ICG fluorescence; **PSMA-radioguided surgery**.
- **Artificial intelligence:** video-based skill assessment, automated performance metrics (kinematic data predicting continence after RARP), and phase recognition.
- **Autonomy:** graded levels (0–5); current systems are level 0–1. Experimental autonomous suturing (STAR) has been done in animals. Image-guided automated **Aquablation** (AquaBeam) is the first semi-autonomous resection in urology, echoing PROBOT.
- **Robotic endourology:** robotic flexible ureteroscopy (e.g. Avicenna Roboflex) and the Monarch platform for PCNL access.

**Diagram to draw:** block diagram of da Vinci (surgeon console linked to vision cart and patient-side cart, with EndoWrist 7 DOF inset); port placement for RARP (6-port transperitoneal fan).

**Key points to remember:** - PUMA 560 (1985) was the first surgical robot. PROBOT (Imperial College) was the first urological robot. da Vinci was FDA-approved in 2000. The first RARP was by Binder & Kramer in 2000. - Yaxley (Lancet 2016) and Coughlin (Lancet Oncol 2018) found no functional difference between open and robotic RP at 12 weeks and 24 months. - RAZOR (Lancet 2018): RARC non-inferior for 2-year PFS. iROC (JAMA 2022): intracorporeal RARC gave more days alive and out of hospital. - Generations: S, Si, Xi (2014), SP (2018), da Vinci 5 (2024). Alternatives: Hugo, Versius, Senhance, SSI Mantra (India). - Steep Trendelenburg causes raised IOP/ICP and ischaemic optic neuropathy; prolonged lithotomy causes compartment syndrome.

## IV-2. Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation

**Type:** LAQ (25 marks) | ~1600 words

## Introduction

- The gap between demand and organ supply has driven wider use of **marginal (extended criteria) donors**. These are donors whose kidneys carry a higher risk of delayed graft function (DGF), primary non-function or shorter graft survival.
- For many wait-listed patients, a marginal kidney still gives better survival than staying on dialysis.
- Campbell: Renal transplantation chapter. Guidelines: KDIGO (candidate evaluation 2020; living donor 2017), EAU Renal Transplantation guideline.

## Part A: Marginal kidney donor

### Definitions and categories

#### 1. Expanded criteria donor (ECD), UNOS 2002 (deceased, brain-dead):

- Any donor **≥60 years**, or
- A donor aged **50-59 years** with **≥2** of the following:
  - history of **hypertension**
  - **terminal serum creatinine >1.5 mg/dL**
  - **death from cerebrovascular accident**
- ECD kidneys carry about a 70% higher relative risk of graft loss than standard criteria donor (SCD) kidneys (RR >1.7).

#### 2. KDRI / KDPI

- **KDRI** (Kidney Donor Risk Index; Rao 2009) is built from 10 donor factors: age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV status and DCD status.
- **KDPI** is the KDRI expressed as a percentile (0-100%).
- In the **US Kidney Allocation System (December 2014)**, **KDPI >85%** replaced the ECD label.
- Under the same system, the **top 20% EPTS** recipients are matched to kidneys with **KDPI ≤20%** ("longevity matching").

#### 3. Donation after circulatory death (DCD): Maastricht classification

| Category | Description                                                         | Type                          |
|----------|---------------------------------------------------------------------|-------------------------------|
| I        | Dead on arrival / found dead                                        | Uncontrolled                  |
| II       | Unsuccessful resuscitation                                          | Uncontrolled                  |
| III      | Awaiting cardiac arrest after withdrawal of life-sustaining therapy | <b>Controlled</b> (commonest) |
| IV       | Cardiac arrest in a brain-dead donor                                | Controlled/uncontrolled       |
| V        | Euthanasia / medically assisted death (added in some countries)     | Controlled                    |

- The **modified Maastricht classification (Paris, 2013)** subdivided the uncontrolled categories.
- Compared with brain-dead donor kidneys, DCD kidneys have **higher DGF rates (roughly 40-50%)** but **similar long-term graft survival**.

#### 4. Other marginal donors

- Donors with **AKI**: graft outcomes are acceptable if renal function was previously normal, and these kidneys should not be discarded on terminal creatinine alone.
- Older, diabetic or hypertensive donors, and donors with proteinuria.
- **HCV-viraemic donor to HCV-negative recipient**, followed by **direct-acting antivirals (DAAs)** with cure rates near 100%. THINKER trial (Goldberg, NEJM 2017).

- **HIV-positive donor to HIV-positive recipient:** permitted in the US by the **HOPE Act (2013)**. South African experience (Muller) came first.
- **Small paediatric donors:** en bloc kidneys (both kidneys with the aorta and IVC) can be transplanted into adults. They carry a higher thrombosis risk but give good long-term function.
- Donors with **treated low-grade malignancy** (e.g. small RCC that has been excised) or those at increased infectious risk.

## 5. Marginal living donors (KDIGO 2017)

- Older donors; donors with controlled hypertension (on 1–2 drugs, no end-organ damage); obesity (assessed individually, with BMI >30–35 needing caution); isolated microscopic haematuria after a negative work-up; minor anatomical variants; single stone episode.
- KDIGO 2017 recommends a **risk-based approach**: each centre sets an acceptable threshold of projected **lifetime ESKD risk**.

## Assessment and optimisation of marginal kidneys

### Donor biopsy: Remuzzi score

- Grades glomerulosclerosis, tubular atrophy, interstitial fibrosis and arterial narrowing, each scored 0–3 (total 0–12).
  - **0–3:** single transplant
  - **4–6:** dual kidney transplant
  - **≥7:** discard
- **Glomerulosclerosis >20%** has traditionally been linked to poorer outcomes.
- Frozen-section biopsy has limitations, and over-reliance on biopsy leads to unnecessary discards.

### Machine perfusion

- **Hypothermic machine perfusion (HMP):**
  - **Moers et al., NEJM 2009** (Eurotransplant RCT, paired kidneys): HMP **reduced DGF** (about 21% vs 27%) and **improved 1-year graft survival**. The benefit was greatest in ECD kidneys.
  - Resistance indices and flow give an additional viability assessment, which should not be used alone.
- **Normothermic machine perfusion (NMP)** (Nicholson, Hosgood; Cambridge, UK) at 37 °C with oxygenated red cells allows viability assessment and repair (a "quality score"). It is still largely within trials.

### Other optimisation measures

- **Short cold ischaemia time:** under 12–18 h is ideal for ECD/DCD kidneys.
- **Dual kidney transplantation (DKT):** both marginal kidneys go into one recipient to supply enough nephron mass. Placement can be unilateral or bilateral in the iliac fossae.
- **Donor management:** e.g. mild donor hypothermia has been studied to reduce DGF.

### Recipient matching

- **"Old-for-old":** the **Eurotransplant Senior Program (ESP, 1999)** allocates kidneys from donors **≥65 years** to recipients **≥65 years**. It prioritises local allocation and short cold ischaemia time over HLA matching.
- Marginal kidneys suit older, diabetic recipients, who have a long expected wait and high dialysis mortality.
- Patients must give **informed consent** to receive an ECD/high-KDPI kidney.

## Outcomes

- **Wolfe et al., NEJM 1999:** transplantation gave a large survival benefit over remaining on the waiting list.
- **Ojo et al. (JASN 2001):** ECD transplantation also improved survival compared with dialysis, especially in recipients over 40 years, diabetics and at centres with long waiting times.
- Graft survival with ECD/high-KDPI kidneys is lower than with SCD kidneys, but patient survival exceeds that of matched patients staying on dialysis.
- **DGF management:** avoid nephrotoxic early high-dose CNI by delaying CNI under antibody induction.

## Part B: Newer immunosuppressant drugs

### Overview of the current standard

- **Induction:** basiliximab or rATG.
- **Maintenance (triple therapy): tacrolimus + mycophenolate + steroids**, with or without early steroid withdrawal.
- "Newer" agents aim to avoid calcineurin inhibitor (CNI) nephrotoxicity, improve adherence, desensitise highly sensitised patients and treat antibody-mediated rejection (AMR).

### Induction agents

| Drug                              | Mechanism                                                                       | Dose (usual)                        | Key points                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Basiliximab</b>                | Chimeric anti-CD25 (IL-2 receptor $\alpha$ ) monoclonal antibody; non-depleting | <b>20 mg IV on day 0 and day 4</b>  | Low risk; well tolerated                                                                   |
| <b>Rabbit ATG (Thymoglobulin)</b> | Polyclonal T-cell depletion                                                     | 1–1.5 mg/kg/day; total ~4.5–6 mg/kg | High immunological risk, DGF. Cytokine release, leucopenia, thrombocytopenia, CMV and PTLD |
| <b>Alemtuzumab</b>                | Anti-CD52 (lymphocytes)                                                         | 30 mg single dose (off-label)       | Profound prolonged depletion; allows steroid/CNI minimisation                              |

### Maintenance agents (newer and modified)

| Drug / class                                                                                             | Mechanism                                                                                                      | Points                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extended-release tacrolimus: Advagraf</b> (once daily), <b>Envarsus XR</b> (LCP-tacrolimus, MeltDose) | Calcineurin inhibition reduces IL-2 transcription                                                              | Once-daily dosing improves adherence. Envarsus has higher bioavailability (lower dose, flatter peak) and fewer tremors                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Mycophenolate</b> (MMF; enteric-coated mycophenolate sodium, EC-MPS)                                  | Inhibits IMPDH, blocking de novo purine synthesis in lymphocytes                                               | GI upset and leucopenia; <b>teratogenic</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>mTOR inhibitors: sirolimus, everolimus</b>                                                            | Bind FKBP12 and block mTOR, arresting the G1 to S cell cycle                                                   | Non-nephrotoxic, <b>antiproliferative</b> with fewer skin cancers, and <b>fewer CMV/BK infections</b> . <b>TRANSFORM study (Pascual, JASN 2018)</b> : everolimus + reduced-exposure CNI was <b>non-inferior</b> to MPA + standard CNI, with less CMV/BK viraemia. Side effects: <b>impaired wound healing, lymphocele</b> , proteinuria, hyperlipidaemia, pneumonitis, mouth ulcers. Avoid early after surgery in obese patients (wound problems)                                                                                               |
| <b>Belatacept</b>                                                                                        | <b>CTLA4-Ig</b> fusion protein that binds <b>CD80/CD86</b> , blocking the <b>CD28 costimulation</b> "signal 2" | FDA 2011. IV infusion: 10 mg/kg (days 1, 5, then weeks 2, 4, 8, 12), then <b>5 mg/kg every 4 weeks</b> . <b>BENEFIT</b> (living/SCD) and <b>BENEFIT-EXT</b> (ECD) trials: <b>better GFR</b> than ciclosporin, but <b>higher early acute rejection</b> . <b>7-year BENEFIT data (Vincenti, NEJM 2016)</b> : improved patient and graft survival. <b>Contraindicated in EBV-seronegative recipients</b> because of the <b>PTLD risk (especially CNS)</b> . Avoids CNI nephrotoxicity, hypertension and diabetes. Monthly infusions help adherence |

## Desensitisation and antibody-mediated rejection

| Drug                                                                     | Mechanism                                                                 | Use                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IVIG</b>                                                              | Immunomodulation, anti-idiotypic effect, FcRn saturation                  | Desensitisation; AMR (with plasmapheresis)                                                                                                                                                                                                                                           |
| <b>Rituximab</b>                                                         | Anti-CD20 B-cell depletion                                                | ABO-incompatible protocols; AMR adjunct                                                                                                                                                                                                                                              |
| <b>Bortezomib</b>                                                        | Proteasome inhibitor causing plasma-cell apoptosis                        | AMR. The <b>BORTEJECT RCT</b> (late AMR) was negative. Neuropathy                                                                                                                                                                                                                    |
| <b>Eculizumab</b>                                                        | Anti-C5, blocking the membrane attack complex                             | <b>aHUS recurrence</b> prevention and treatment; selected early AMR. Meningococcal vaccination is mandatory                                                                                                                                                                          |
| <b>Imlifidase (IdeS)</b>                                                 | <i>Streptococcus pyogenes</i> enzyme that <b>cleaves IgG</b> within hours | <b>EMA conditional approval 2020</b> to desensitise <b>highly sensitised</b> patients with a positive crossmatch to a deceased donor. A single dose of 0.25 mg/kg. Antibody rebound occurs, and ATG/rituximab must be given after imlifidase because it also cleaves therapeutic IgG |
| <b>Tocilizumab</b> (anti-IL-6 receptor), <b>clazakizumab</b> (anti-IL-6) | Block the IL-6 pathway that drives DSA and inflammation                   | <b>Chronic active AMR</b> : investigational, not approved                                                                                                                                                                                                                            |
| <b>Felzartamab</b>                                                       | Anti-CD38 (plasma cells and NK cells)                                     | Phase 2 trial in late AMR (NEJM 2024) showed promising effects; under study                                                                                                                                                                                                          |
| <b>C1-esterase inhibitor</b>                                             | Complement inhibition                                                     | AMR trials                                                                                                                                                                                                                                                                           |

### Caution about voclosporin

- Voclosporin is a novel CNI that is **approved for lupus nephritis (2021)**.
- It is **not approved for renal transplantation**; earlier transplant studies did not lead to registration.

### Tolerance and the future

- Tolerance induction**: combined kidney and haematopoietic stem-cell transplantation (mixed chimerism) from HLA-matched or haploidentical living donors (Stanford, Northwestern and MGH protocols). Selected patients have been able to stop immunosuppression.
- Regulatory T-cell (Treg) therapy**: ONE Study and TWO Study.
- Anti-CD40/CD40L agents** (e.g. tegoprubart): investigational, and also used in xenotransplantation.
- Xenotransplantation**: gene-edited pig kidney into a living human recipient (MGH, March 2024), followed by further compassionate-use cases and trials.

### Side-effect summary

| Drug                 | Signature toxicity                                                          |
|----------------------|-----------------------------------------------------------------------------|
| Tacrolimus           | Nephrotoxicity, NODAT, tremor, hypertension, hyperkalaemia, hypomagnesaemia |
| MMF                  | GI upset, cytopenia, teratogenicity                                         |
| mTOR inhibitors      | Wound healing problems, lymphocele, proteinuria, dyslipidaemia, pneumonitis |
| Belatacept           | PTLD (EBV-), early rejection                                                |
| Depleting antibodies | Infection (CMV), PTLD, cytokine release                                     |

## Recent advances

- Normothermic perfusion with ex vivo repair.
- Using HCV-positive kidneys with DAAs.
- Using **imlifidase** for highly sensitised patients.
- **Donor-derived cell-free DNA** and molecular microscope (MMDx) diagnostics for rejection.
- Pig-kidney xenotransplantation.

**Diagram to draw:** three-signal model of T-cell activation (signal 1 TCR-MHC, blocked by ATG; signal 2 CD28-CD80/86, blocked by belatacept; signal 3 IL-2R-mTOR, blocked by basiliximab and sirolimus/everolimus), with the CNi and MMF sites marked.

**Key points to remember:** - ECD (UNOS 2002): age  $\geq 60$ , or 50-59 with  $\geq 2$  of HTN, creatinine  $>1.5$  mg/dL, CVA death. Replaced by KDPI  $>85\%$  in December 2014. - Remuzzi score: 0-3 single, 4-6 dual,  $\geq 7$  discard. HMP reduces DGF (Moers, NEJM 2009). Eurotransplant Senior Program:  $\geq 65$  to  $\geq 65$ . - Belatacept (CTLA4-Ig): BENEFIT/BENEFIT-EXT show better GFR. Contraindicated in EBV-seronegative recipients (PTLD). - TRANSFORM: everolimus + low-dose CNi is non-inferior, with fewer CMV/BK infections. - Imlifidase (IgG-cleaving enzyme) has EMA conditional approval (2020) for highly sensitised recipients. Voclosporin is for lupus nephritis, not transplantation.

## IV-3a. Etiology and management of hypospadias fistula

**Type:** SAQ (10 marks) | ~550 words

### Definition and incidence

- A **urethrocutaneous fistula (UCF)** is an abnormal communication between the neourethra and the penile skin. It is the **commonest complication of hypospadias repair**.
- Incidence:
  - **Distal TIP (Snodgrass):** about  **$<5-10\%$** .
  - **Proximal and staged repairs:** higher, often  **$10-25\%$**  or more.
  - Highest in **redo** repairs.
- Most fistulas appear in the early postoperative period. Some present late, so long-term follow-up is needed.
- Campbell: Hypospadias chapter (paediatric).

### Aetiology (risk factors)

#### Local or technical factors

- **Distal obstruction:** meatal stenosis, neourethral stricture, or a narrow neourethra causing high voiding pressure. This is the **most important correctable cause**.
- **Ischaemia** from poor flap vascularity, excess diathermy or tight dressing.
- **Infection** and **haematoma**.
- **Tension** on the suture line, and **overlapping suture lines** (urethral and skin lines on top of each other).
- **No barrier (interposition) layer** between the neourethra and the skin.
- Poor-quality tissue: a small glans, a shallow urethral plate, previous surgery, or a deficient spongiosum.
- Suture material: subcuticular, inverting, absorbable fine suture (6-0/7-0 PGA) is preferred.

#### Patient factors

- Proximal hypospadias, redo surgery, and small glans size (under about 14 mm). The role of age and preoperative androgens is debated.
- **Catheter or stent problems:** early removal, blockage or **bladder spasm**.

## Prevention

- **Barrier layer:**
  - **dorsal dartos (preputial) pedicled flap**
  - **tunica vaginalis flap** (especially for proximal or redo repairs)
  - spongioplasty
- These layers reduce fistula rates significantly.
- Other measures:
  - non-overlapping suture lines, with a tension-free glanuloplasty
  - meticulous haemostasis and magnification
  - adequate urinary drainage

## Evaluation

- Examine during voiding: site, number and size. A coronal fistula may mean glans dehiscence.
- **Rule out distal obstruction:**
  - calibrate the meatus and neourethra
  - urethroscopy/cystoscopy (often at the time of repair)
  - uroflowmetry
- Saline or methylene-blue injection through the meatus at surgery (with the urethra compressed proximally) identifies small or multiple fistulas.
- Skin cover: is there enough local dartos?

## Management

- **Timing: wait at least 6 months** after the primary repair for inflammation to settle and tissue to mature. A few tiny early fistulas close on their own.
- **Always correct distal obstruction** (meatotomy, meatoplasty, stricture management) at the same sitting. Otherwise the fistula recurs.

| Fistula type                                       | Procedure                                                                                                                                                                 |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Small (&lt;2-3 mm), single, away from glans</b> | Excise the tract, close in layers (inverting subcuticular 6-0/7-0 absorbable), add a <b>dartos/subcutaneous barrier flap</b> , rotate skin so suture lines do not overlap |
| <b>Large, multiple, or with a thin urethra</b>     | <b>Redo urethroplasty</b> (TIP redo, onlay flap, or staged BMG)                                                                                                           |
| <b>Coronal / subcoronal (glans dehiscence)</b>     | Re-do glanuloplasty or meatoplasty (redo TIP), rather than simple closure                                                                                                 |
| <b>Barrier options</b>                             | Local dartos, <b>tunica vaginalis flap</b> , de-epithelialised skin flap; Y-V skin advancement for skin cover                                                             |

- **Diversion:** small simple closures can be left without a catheter or with a short-term stent. Larger or redo repairs need a stent or catheter for 5–10 days.
- **Success:** simple layered closure succeeds in about **80-90%**. Results are worse for large, multiple or recurrent fistulas.
- **Recurrence:** look for missed distal obstruction.

## Recent advances

- Routine use of **barrier layers** and **tunica vaginalis** flaps in proximal repairs.

- **Tissue sealants** (fibrin glue) and buccal grafts in staged redo surgery.
- Validated outcome scores (**HOSE, HOPE**) for standardised reporting of complications.

UCF → wait ≥6 mo → calibrate/cystoscopy  
 ↳ distal obstruction? → correct same sitting  
 ↳ small single → excise + layered closure + dartos/TV flap  
 ↳ large/multiple/glans dehiscence → redo urethroplasty

**Diagram to draw:** cross-section of the repair showing the neourethra, dartos barrier flap and skin, with suture lines offset.

**Key points to remember:** - Fistula is the commonest complication. TIP distal rate is under 5–10%; proximal repairs carry a higher rate. - Distal obstruction (meatal stenosis) must always be excluded and corrected. - Wait at least 6 months, then do a layered closure with a barrier (dartos or tunica vaginalis) and non-overlapping suture lines. - Coronal fistula equals glans dehiscence and needs redo glanuloplasty.

## IV-3b. Silodosin

**Type:** SAQ (10 marks) | ~550 words

### Definition

- Silodosin is a **highly uroselective α1A-adrenoceptor antagonist** (indoline derivative), used mainly for **LUTS due to BPH**.
- Campbell: Medical management of BPH chapter. EAU Non-neurogenic male LUTS guideline. AUA BPH guideline.

### Pharmacology

- **Receptor selectivity:**
  - α1A : α1B about **162 : 1**
  - α1A : α1D about **55 : 1**
- Practical consequence: very little effect on vascular α1B receptors, so **orthostatic hypotension is minimal**. The α1A blockade in the prostate, bladder neck, urethra and seminal tract causes **ejaculatory dysfunction**.

### Pharmacokinetics

- Oral bioavailability about **32%**. It is taken **with a meal**.
- Metabolised by **CYP3A4, UGT2B7** (forming the active glucuronide KMD-3213G, which has a long half-life) and alcohol/aldehyde dehydrogenase.
- It is a **P-glycoprotein** substrate.
- Excreted in urine and faeces.

## Dose and contraindications

| Item             | Detail                                                                                                                                                                                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dose             | <b>8 mg once daily with a meal</b> (4 mg capsules also available)                                                                                                                                |
| Renal impairment | <b>Moderate (CrCl 30-50 mL/min): 4 mg daily</b>                                                                                                                                                  |
| Contraindicated  | <b>Severe renal impairment (CrCl &lt;30), severe hepatic impairment (Child-Pugh C), strong CYP3A4 inhibitors</b> (ketoconazole, clarithromycin, ritonavir), strong P-gp inhibitors (cyclosporin) |
| Caution          | Other $\alpha$ -blockers, PDE5 inhibitors (minor additive BP effect), planned cataract surgery                                                                                                   |

## Efficacy

- **US phase III RCTs (Marks et al., 2009):** compared with placebo, silodosin improved **IPSS and Qmax**, with onset within **2-4 days** (Qmax improved within hours of the first dose).
- **Chapple et al., Eur Urol 2011** (European RCT, silodosin 8 mg vs tamsulosin 0.4 mg vs placebo, 12 weeks):
  - silodosin was **non-inferior to tamsulosin** for IPSS
  - only silodosin significantly reduced **nocturia** compared with placebo
  - abnormal ejaculation was more frequent with silodosin
- Long-term open-label extensions show sustained efficacy.
- Silodosin does not reduce prostate volume or the risk of acute urinary retention or surgery. Combine it with a 5-ARI for large glands.

## Adverse effects

- **Retrograde ejaculation / anejaculation:** about **28%** in US trials. This is the **most common** side effect and is reversible.
  - Mechanism: reduced seminal emission (vas, seminal vesicles) rather than true retrograde flow.
- Dizziness, orthostatic hypotension (<3%), nasal congestion, diarrhoea, headache.
- **Intraoperative floppy iris syndrome (IFIS):** warn the patient and inform the ophthalmologist before cataract surgery.

## Uses

1. **BPH/LUTS:** first-line  $\alpha$ -blocker. Useful in elderly and cardiac patients because of low hypotension risk.
2. **Medical expulsive therapy (MET), off-label:** for **distal ureteric stones >5 mm**.
  - Meta-analyses and small RCTs suggest silodosin may give expulsion rates similar to or higher than tamsulosin.
  - Large tamsulosin RCTs (**SUSPEND, Lancet 2015**) showed no benefit overall.
  - **EAU** supports  $\alpha$ -blockers for distal stones >5 mm.
3. **Ureteric stent-related symptoms:** improves pain and urinary scores.
4. **Chronic prostatitis / chronic pelvic pain syndrome:** as part of multimodal UPOINT therapy, especially in  $\alpha$ -blocker-naïve men.
5. Also used for premature ejaculation (exploiting the anejaculation effect) and to facilitate ureteral access sheath placement (studied).

## Comparison of $\alpha$ -blockers

| Drug              | Selectivity                             | Dose                    | Titration           | Orthostatic hypotension       | Ejaculatory dysfunction |
|-------------------|-----------------------------------------|-------------------------|---------------------|-------------------------------|-------------------------|
| <b>Silodosin</b>  | $\alpha$ 1A (very high)                 | 8 mg OD                 | No                  | Least                         | <b>Highest (~28%)</b>   |
| <b>Tamsulosin</b> | $\alpha$ 1A = $\alpha$ 1D > $\alpha$ 1B | 0.4 mg OD               | No                  | Low                           | Moderate (~10%)         |
| <b>Alfuzosin</b>  | Non-selective (clinically uroselective) | 10 mg OD (XL)           | No                  | Low-moderate                  | Least                   |
| <b>Doxazosin</b>  | Non-selective                           | 1-8 mg OD (GITS 4-8 mg) | Yes (standard form) | Higher                        | Low                     |
| <b>Terazosin</b>  | Non-selective                           | 1-10 mg OD at night     | Yes                 | Highest ("first-dose" effect) | Low                     |

## Recent advances

- Fixed-dose combinations with dutasteride or tadalafil (in some markets).
- Use of silodosin in MET and stent symptoms.
- Interest in silodosin for premature ejaculation.

**Diagram to draw:**  $\alpha$ 1 receptor subtype distribution ( $\alpha$ 1A in prostate/bladder neck,  $\alpha$ 1B in vessels,  $\alpha$ 1D in bladder/spinal cord) with the action of silodosin marked.

**Key points to remember:** - Selectivity  $\alpha$ 1A :  $\alpha$ 1B is about 162 : 1. Dose 8 mg OD with a meal; 4 mg if CrCl 30-50; contraindicated if CrCl <30, Child-Pugh C, or with strong CYP3A4 inhibitors. - Commonest adverse effect is anejaculation/retrograde ejaculation (~28%). Lowest hypotension risk of the  $\alpha$ -blockers. - Chapple 2011: non-inferior to tamsulosin and also reduced nocturia. - Warn about IFIS before cataract surgery.

## IV-3c. Sacral neuromodulation

**Type:** SAQ (10 marks)

**See Paper III, Q3a (Sacral neuromodulation): identical question.**

**Recent advances angle (emphasise in a Recent Advances paper):** - **Rechargeable, miniaturised IPGs:** - **Axonics r-SNM** (FDA 2019; about 15-year life) - **Medtronic InterStim Micro** (rechargeable) - recharge-free long-life devices (**InterStim X**, **Axonics F15**) - **Full-body MRI-conditional** systems (1.5 T and 3 T with SureScan/compatible leads), which solve the main historical limitation of SNM. - **Standardised tined-lead placement** with curved stylet and fluoroscopic landmarks (Matzel), improving test-phase success and reducing reoperation. - **Closed-loop / sensing SNM** using evoked compound action potential (ECAP) recordings to auto-adjust stimulation is at the research stage. - **Implantable tibial nerve stimulators:** **eCoin** (FDA 2022), **Revi** (BlueWind, FDA 2023) give less-invasive alternatives for refractory OAB/UUI.

## IV-3d. Urethral stents

**Type:** SAQ (10 marks) | ~550 words

## Definition

- A urethral stent is a tubular endoprosthesis placed endoscopically in the **prostatic or bulbar urethra** to keep the lumen open. It is a **minimally invasive alternative** to surgery in men unfit for definitive treatment.
- Campbell: Urethral stricture / reconstruction and BPH minimally invasive therapy chapters.

## Classification

| Type                                  | Examples                                                                                                                                                                                                                                                    | Features                                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Temporary, non-epithelialising</b> | <b>Memokath</b> (thermo-expandable <b>nickel-titanium</b> shape-memory spiral; e.g. Memokath 028 prostatic, 044 bulbar), <b>Allium</b> (covered, self-expanding nitinol; BUS bulbar and prostatic stents), polyurethane / Spanner temporary prostatic stent | Do not become epithelialised; removable; need replacement or removal over months to years                    |
| <b>Permanent, epithelialising</b>     | <b>UroLume</b> (woven superalloy mesh)                                                                                                                                                                                                                      | Covered by urothelium in 3–6 months; difficult removal.<br><b>UroLume has been withdrawn from the market</b> |
| <b>Biodegradable</b>                  | PLGA spiral stents (experimental)                                                                                                                                                                                                                           | Dissolve; avoid removal                                                                                      |
| By site                               | Prostatic (BPH, retention) vs <b>bulbar urethral</b> (stricture) vs <b>sphincteric</b> (DSD)                                                                                                                                                                |                                                                                                              |

## Indications

1. **BPH with retention or LUTS in high-risk patients** unfit for TURP or anaesthesia, as an alternative to a long-term catheter.
2. **Recurrent bulbar urethral stricture** in men **unfit for or unwilling to undergo urethroplasty**. Short bulbar strictures only; avoid in the penile urethra (pain, erection problems).
3. **Detrusor-sphincter dyssynergia (DSD) in spinal cord injury**: sphincteric stent (UroLume, Memokath) as a reversible alternative to **external sphincterotomy**.
4. Others: palliation of obstruction from prostate cancer, post-brachytherapy and post-radiation stenosis (limited).

## Technique

- Done under local or regional anaesthesia: urethroscopy, measurement of stricture length, placement under endoscopic or fluoroscopic guidance.
- Memokath expands at about 55 °C and softens when cooled for removal.
- The stent must not cross the external sphincter in a stricture patient, to avoid incontinence. Bulbar stents start distal to the sphincter.

## Complications

| Complication                              | Comment                                                                        |
|-------------------------------------------|--------------------------------------------------------------------------------|
| Migration                                 | Commonest with temporary stents                                                |
| Encrustation, stones                      | Especially in prostatic stents                                                 |
| Hyperplastic tissue ingrowth / restenosis | Major problem with UroLume (end ingrowth); this needs resection                |
| UTI, haematuria                           |                                                                                |
| Incontinence, post-void dribbling         | Especially if the stent crosses the sphincter                                  |
| Perineal pain, painful erections          | Bulbar stents                                                                  |
| Difficult removal                         | Epithelialised stents; may need open urethroplasty, which becomes more complex |

## Results

- **Prostatic stents** in unfit men: most become catheter-free in the short term, but a significant proportion need removal for migration, encrustation or symptoms.
- **UroLume for strictures:**
  - early results were good
  - long-term follow-up showed high rates of **recurrent obstruction from tissue hyperplasia**
  - the need for repeat procedures, and the difficult salvage urethroplasty, led to its abandonment
- **DSD sphincteric stents:** comparable to sphincterotomy in the short term. Later migration and ingrowth are common.

## Place in current guidelines

- **AUA Urethral Stricture guideline (2016, amended 2023): urethral stents should not be placed in patients who are considering urethroplasty.** A stent makes later reconstruction harder.
- **EAU Urethral Stricture guideline:** stents are **not recommended as routine** therapy. They are reserved for selected unfit patients.
- Urethroplasty remains the standard for recurrent strictures.

## Recent advances

- **Optilume drug-coated balloon** (paclitaxel-coated urethral balloon; FDA 2021) for recurrent anterior strictures of 3 cm or less.
  - **ROBUST III RCT (Elliott, J Urol 2022):** much higher stricture-free rates than DVIU or dilation at 6 months (about 75% vs 27%).
  - It is a balloon, not a stent.
- **iTIND** (temporary implantable nitinol device; FDA 2020): placed in the prostatic urethra for 5–7 days, remodelling the bladder neck and prostatic urethra by ischaemic incision, then removed. It is a MIST option for BPH.
- **Zenflow Spring:** an investigational prostatic implant for BPH.
- **Allium** covered stents and biodegradable stents are under evaluation.

**Diagram to draw:** sagittal urethra showing a prostatic stent positioned between the bladder neck and verumontanum, and a bulbar stent distal to the external sphincter.

**Key points to remember:** - Two types: temporary (Memokath nitinol thermo-expandable, Allium) and permanent epithelialising (UroLume, now withdrawn). - Indications: unfit BPH patients, unfit bulbar stricture patients, DSD as an alternative to sphincterotomy. - Main

problems: migration, encrustation, hyperplastic ingrowth. - AUA: do not place a stent in a man considering urethroplasty. - Optilume (ROBUST III) is the recent advance for recurrent short strictures.

## IV-3e. Studer Neobladder

Type: SAQ (10 marks) | ~550 words

### Introduction

- The Studer neobladder is an **orthotopic ileal low-pressure reservoir** anastomosed to the urethra. It was developed by **Urs E. Studer (Bern, Switzerland)** in the mid-1980s and first published in the late 1980s.
- Campbell: Orthotopic urinary diversion chapter.

### Principles

- Detubularisation**: interrupts coordinated peristaltic contractions, giving **low pressure**.
- Spherical reconfiguration** (Laplace's law,  $P = 2T/r$ ): the largest volume for the surface area gives **high capacity at low pressure**.
- An **isoperistaltic afferent tubular limb** works as an antireflux mechanism and allows a tension-free ureteric anastomosis with ureters of different lengths.
- The pouch lies orthotopically and the patient **voids by abdominal straining (Valsalva) with pelvic floor relaxation**, maintaining continence through the intact **rhabdosphincter**.

### Technique (Campbell)

- Isolate **54-56 cm of terminal ileum**, ending about **25 cm proximal to the ileocaecal valve** (to preserve bile-salt and B12 absorption).
- Restore bowel continuity.
- Distal ~40-44 cm**: opened along the **antimesenteric border**, arranged in a **U**. The medial borders are sutured to form the back wall.
- Proximal ~12-14 cm**: left **tubular** as the **isoperistaltic afferent (chimney) limb**.
- Ureteroileal anastomosis**: end-to-side, refluxing (Nesbit technique) of each ureter onto the afferent limb, over stents (Wallace anastomosis is an alternative).
- Fold the U-shaped plate (the bottom of the U folded up between the arms) to make a **spherical reservoir**. The **most dependent point** gets a small opening for the **urethral anastomosis** (6 sutures), tension-free.
- Stents and a suprapubic/urethral catheter go in. The initial capacity is about 120-150 mL, increasing to **400-500 mL** over months.

### Indications and contraindications

- Indications**: radical cystectomy (MIBC, high-risk NMIBC) in a motivated patient with a **negative urethral margin** (frozen section at the apical urethra), and in selected women (bladder neck free of tumour).
- Contraindications**: **impaired renal function** (commonly quoted thresholds: eGFR below ~35-50 mL/min or serum creatinine >2 mg/dL), **hepatic dysfunction**, **positive urethral margin**, IBD/short bowel/radiation enteritis, inability to perform **CISC** or poor compliance, incompetent external sphincter.

## Postoperative care

- Daily catheter **irrigation** for mucus; **sodium bicarbonate** for acidosis; catheter out after a cystogram (~10–14 days).
- **Voiding training**: timed voiding every 2 h, extended to 3–4 h; sit, relax the pelvic floor and use Valsalva.
- **Nocturnal alarm** (void once or twice at night) to prevent enuresis and overdistension. Pelvic floor exercises.

## Outcomes

- **Daytime continence: about 90%** at 1 year.
- **Night-time continence: about 70–80%**. This improves over time but declines with age.
- **CISC** needed in a minority of men (under 10–20%). The need is **higher in women** (hypercontinence, which can be up to 20–40% in some series).

## Complications

| Early                                  | Late                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------|
| Urine leak, ileus, UTI, pyelonephritis | <b>Hyperchloraemic metabolic acidosis</b> (ileal absorption of $\text{NH}_4\text{Cl}$ ) |
| Mucus retention                        | <b>Vitamin B12 deficiency</b> (start screening at 3–5 years)                            |
|                                        | <b>Ureteroileal anastomotic stricture</b> (~3–10%)                                      |
|                                        | <b>Stones</b> (in pouch: mucus, staples, infection)                                     |
|                                        | <b>Incontinence</b> (day/night) or <b>hypercontinence/retention</b>                     |
|                                        | Pouch–vaginal fistula (women); urethral recurrence; bowel dysfunction (diarrhoea)       |

## Comparison with other orthotopic neobladders

| Neobladder                                          | Key features                                                                                                                     |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Studer</b>                                       | U-shaped detubularised pouch with an isoperistaltic <b>afferent limb</b> , refluxing ureteric anastomosis                        |
| <b>Hautmann (W)</b>                                 | Longer ileal segment (~60–70 cm) folded into a <b>W</b> ; larger initial capacity; ureters implanted directly (or with chimneys) |
| <b>Robotic intracorporeal (Karolinska, Wiklund)</b> | Studer-type pouch built intracorporeally                                                                                         |

## Recent advances

- Fully **intracorporeal robotic Studer neobladder**; **ERAS** pathways.
- Nerve-sparing and pelvic-organ-sparing cystectomy in women to reduce hypercontinence.

**Diagram to draw:** (1) 54–56 cm ileum marked into a 12–14 cm afferent limb and 40–44 cm detubularised segment, 25 cm from the ileocaecal valve. (2) The final spherical pouch with the afferent limb receiving both ureters and the urethral anastomosis at the most dependent point.

**Key points to remember:** - Segment: 54–56 cm of ileum ending 25 cm proximal to the ICV. Distal 40–44 cm detubularised into a U; proximal 12–14 cm isoperistaltic afferent limb. - Principles: detubularisation plus spherical shape (Laplace) gives a low-pressure, high-capacity reservoir. - Continence about 90% by day and 70–80% at night. Women have more hypercontinence and need CISC more often. - Main metabolic issues: hyperchloraemic acidosis and B12 deficiency. Avoid with significantly impaired renal function (eGFR below ~35–50 mL/min).

## IV-3f. Cryotherapy for carcinoma prostate

**Type:** SAQ (10 marks) | ~600 words

### Definition

- Cryotherapy (cryoablation, cryosurgical ablation of the prostate, CSAP) is **image-guided transperineal freezing** of prostate tissue to cause **cell death**.
- Campbell: Ablative therapies for prostate cancer chapter (Focal therapy and cryotherapy).

### Physical principle

- **Joule-Thomson effect:** a high-pressure gas expanding through a narrow orifice changes temperature.
  - **Argon** produces freezing to about **−185 °C** at the needle tip.
  - **Helium** produces warming and is used for **active thawing**.
- **Thermocouples** (apex, rectal wall, external sphincter, neurovascular bundles) monitor temperatures in real time.
- A **urethral warming catheter** protects the urethra from sloughing.

### Generations

| Generation             | Features                                                                                               |
|------------------------|--------------------------------------------------------------------------------------------------------|
| <b>1st</b> (1960s–70s) | <b>Liquid nitrogen</b> , large probes, no imaging; high fistula/incontinence                           |
| <b>2nd</b> (1990s)     | <b>TRUS guidance</b> , <b>urethral warmer</b> , thermocouples (Onik)                                   |
| <b>3rd</b>             | <b>Argon/helium</b> , <b>thin 17-gauge needles</b> , template, multiple thermocouples                  |
| <b>4th</b>             | Computer planning, precise ice-ball shaping; enables <b>focal / hemi-gland</b> therapy with MRI fusion |

### Mechanisms of cell death

1. **Direct cellular injury: intracellular ice** (rapid freezing) ruptures membranes; extracellular ice causes osmotic dehydration.
  2. **Vascular injury:** endothelial damage, platelet aggregation and microthrombosis lead to **ischaemic necrosis**. This is the main effect at the margins.
  3. **Apoptosis** at the periphery of the ice ball.
  4. **Immunological effect:** release of tumour antigens may produce a "cryo-immunologic" response.
- **Critical parameters:** target **−40 °C**; **rapid freeze, slow thaw; two freeze-thaw cycles** (the second enlarges the lethal zone).
  - On TRUS the ice-ball edge (~0 °C) is a hyperechoic rim; the lethal isotherm lies inside it.

### Indications

1. **Salvage cryoablation after radiation failure:** the **most important current role**.
  - Requires biopsy-proven local recurrence, no metastases on PSMA PET/CT or MRI, and preferably PSA <10 ng/mL.
2. **Primary whole-gland therapy:** low- to intermediate-risk disease in men unfit or unwilling for RP/RT, or with a small gland (<40–50 mL; larger glands need ADT downsizing).
  - **AUA** lists whole-gland cryosurgery as an option. **EAU** does not recommend it as standard.
3. **Focal / hemi-gland therapy:** MRI-visible unilateral intermediate-risk lesion, within a registry or trial.

## Technique

- TRUS-guided transperineal template placement of cryoneedles, thermocouples and urethral warmer; freeze anterior to posterior to  $-40^{\circ}\text{C}$ , helium thaw, repeat cycle.

## Outcomes

- Low PSA nadir predicts success; failure is usually judged by the **Phoenix definition (nadir + 2 ng/mL)** (not validated specifically for cryotherapy).
- **Donnelly et al., Cancer 2010** (Canadian RCT, cryoablation vs external beam RT): cryotherapy was **non-inferior** for biochemical progression, with fewer positive 36-month biopsies but worse sexual function.
- **Salvage cryotherapy**: 5-year biochemical disease-free survival is reported at roughly 50–60% in selected patients.

## Complications

| Complication                                   | Primary whole gland                | Salvage   |
|------------------------------------------------|------------------------------------|-----------|
| <b>Erectile dysfunction</b>                    | ~ <b>80-90%</b> (focal: much less) | Very high |
| Urinary incontinence                           | ~1-5%                              | ~5-20%    |
| <b>Rectourethral fistula</b>                   | <b>&lt;0.5-1%</b>                  | ~1-3%     |
| Urethral sloughing                             | <5% with a warmer                  | Higher    |
| Urinary retention                              | ~2-5%                              |           |
| Perineal pain, penile numbness, scrotal oedema | Common, transient                  |           |

## Comparison with other ablative energies

|        | Cryo                       | HIFU                                              | IRE                                                   | Focal laser (FLA)           |
|--------|----------------------------|---------------------------------------------------|-------------------------------------------------------|-----------------------------|
| Energy | Freezing (Joule-Thomson)   | Focused ultrasound heat ( $>60^{\circ}\text{C}$ ) | Electric pulses $\rightarrow$ nanopores (non-thermal) | Thermal, MRI-guided         |
| Access | Transperineal              | Transrectal                                       | Transperineal                                         | Transperineal / transrectal |
| Notes  | Anterior tumours treatable | AP height limit ( $\sim 4\text{ cm}$ )            | Spares vessels/nerves                                 | Small lesions               |

## Guideline positions

- **EAU 2024/2025**: primary whole-gland or focal cryotherapy/HIFU **only within trials or registries**. Salvage cryotherapy is an option for selected local recurrence after RT in experienced centres.
- **AUA/ASTRO 2022**: whole-gland cryosurgery may be considered for low/intermediate-risk disease. Focal therapy carries the caveat of limited long-term data.
- **NCCN**: cryotherapy is listed among **local salvage options** after RT failure.

## Recent advances

- MRI-TRUS fusion **focal/hemi-gland cryotherapy**; **PSMA PET** to select salvage patients; cryo plus immunotherapy under study.

**Diagram to draw:** sagittal prostate with transperineal cryoneedles, urethral warmer, thermocouples at the apex and rectal wall, and the ice ball with the  $-40^{\circ}\text{C}$  isotherm inside the  $0^{\circ}\text{C}$  edge.

**Key points to remember:** - Argon freezes and helium thaws (Joule-Thomson). Target  $-40^{\circ}\text{C}$ , with two freeze-thaw cycles. - Cell death comes from intracellular ice, vascular ischaemia (main), apoptosis and an immunological effect. - The main role is salvage after radiation

failure. ED is about 80–90% after whole-gland treatment. Rectourethral fistula is under 1% in primary treatment and higher in salvage. - EAU: primary or focal cryotherapy only within trials or registries. Phoenix definition = nadir + 2.